

A Generalized Vacuum- Λ FLRW Solution and the Sign Convention of k

1. Setup

Starting line element:

$$ds^2 = A(t,r) dt^2 - a(t)^2 B(r) dr^2 - a(t)^2 C(r) r^2 d\theta^2 - a(t)^2 C(r) r^2 \sin^2(\theta) d\phi^2$$

Computing the Ricci tensor (independently verified via direct Christoffel/Riemann summation, no tensor packages) gives four non-zero components R_{00} , R_{01} , R_{11} , R_{22} .

R_{01} gives:

$$R_{01} \propto (da/dt) (\partial A/\partial r) / A = 0$$

Since $da/dt \neq 0$ (non-static universe), this forces $\partial A/\partial r = 0$, i.e. $A(t,r) = A(t)$, confirmed independently.

2. Reduced field equations

With $A=A(t)$, the system reduces to three equations (R_{00} , R_{11} , R_{22}) in three unknowns $A(t)$, $B(r)$, $C(r)$, satisfying $R_{\mu\nu} = \Lambda g_{\mu\nu}$ (vacuum + cosmological constant).

3. Solving R_{00} for $A(t)$

Solving R_{00} for $A(t)$ introduces an integration constant k . The correct general solution (confirmed by direct symbolic substitution) is:

$$A(t) = 3 (da/dt)^2 / (\Lambda a(t)^2 + 3|k|)$$

The absolute value is essential — without it, the formula only reproduces the correct on-shell behavior for one sign of k . The two single-sign candidates,

- $3(da/dt)^2/(\Lambda a^2 - 3k)$ ("standard" form, valid for $k \leq 0$)
- $3(da/dt)^2/(\Lambda a^2 + 3k)$ (valid for $k \geq 0$)

are each individually consistent with R_{00} reducing to $0=0$ — but that check alone is weak: R_{00} alone does not uniquely fix $A(t)$ for an arbitrary $a(t)$. It only closes once $a(t)$ additionally satisfies its own first-integral (Friedmann) constraint, and that constraint's sign is exactly what differs between the two candidates. Two formulas that are mirror images under $k \rightarrow -k$ will always both individually "check out" this way — that is not yet evidence they describe different physics.

4. Solving for B(r), C(r): the general solution

Plugging A(t) (with |k|) into R11 and R22 and solving algebraically:

$$B(r) = (r C'(r) + 2 C(r))^2 / [4 C(r) (r^2 |k| C(r) + 1)]$$

with C(r) left as a completely free function.

This was verified directly: substituting A(t) (with |k|) and this B(r) (with |k|) into both R^1_1 and R^2_2 , with C(r) kept symbolic, both equations reduce to a pure constant identity — independent of r and of the choice of C(r). This is a genuine, exactly verified solution of the field equations for arbitrary C(r), not a special case.

Full solution:

- $A(t,r) = 3(da/dt)^2 / (\Lambda a(t)^2 + 3|k|)$
- $B(t,r) = a(t)^2 (rC' + 2C)^2 / [4C(r^2|k|C + 1)]$
- $\mathcal{N}(t,r) \equiv$ angular metric factor $= a(t)^2 C(r)$
- $k \in \mathbb{R}$ free; C(r) free

This reduces to the standard FLRW metric when $C(r)=1$, $k \leq 0$. For $k>0$, the $C(r)=1$ reduction gives $B(r) = 1/(1+kr^2)$, differing from the standard RCPC form $B(r)=1/(1-kr^2)$.

5. Resolving the $k>0$ discrepancy: curvature invariant check

To settle whether $B(r)=1/(1+kr^2)$ (this work) and $B(r)=1/(1-kr^2)$ (standard) for the *same* k represent different physics or merely different conventions, the intrinsic Ricci scalar of the spatial 3-metric ($C=1$ in both cases — a coordinate-invariant quantity, unaffected by any remaining coordinate freedom) was computed directly:

- Standard metric ($B = 1/(1-kr^2)$): $R_3 = +6k$
- This work's metric ($B = 1/(1+kr^2)$): $R_3 = -6k$

Since the Ricci scalar is a curvature invariant, this is a rigorous, coordinate-independent proof: a solution labeled "k" in this work has the *same actual spatial curvature* as the standard solution labeled "-k". The two families are not different physics — they are the same solution space under the relabeling $k \rightarrow -k$.

6. Conclusion

- The generalized solution with free C(r) is a correctly-derived, independently-verified exact solution of the vacuum+ Λ field equations.
- It is **not** evidence of an error in the standard FLRW equations.
- The apparent disagreement at $k>0$ is fully accounted for by an internal sign convention: this work's k is the negative of the standard curvature parameter.

- The genuinely useful/interesting result is the explicit demonstration — via a fully worked, independently checked exact solution — that the sign of the FLRW curvature parameter k is a convention, not a physically forced choice, together with an explicit free function $C(r)$ generalizing the radial gauge.

Appendix: Notes on conventions used

- Metric signature: $(+,-,-,-)$, with $R_{\mu\nu} = \Lambda g_{\mu\nu}$ as the vacuum+ Λ field equation form used in the original derivation.
- All Ricci tensor computations were performed via direct Christoffel symbol and Riemann tensor summation (brute-force, no tensor-algebra package), cross-checked symbolically at each stage.